Matrix Algebra

IF-1 Let

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 2 \\ -3 & 4 \\ 1 & 4 \end{pmatrix}.$$

Compute:

(a) $B + C,\ B - C,\ 2B - 3C$

(b) $AB,\ AC,\ CA,\ BC^T,\ CB^T$

(c) $A(B + C),\ AB + AC,\ (B + C)A,\ BA + CA$

IF-2 Let A be an arbitrary $m \times n$ matrix and let I_k be the identity matrix of size k. Verify that

$$I_m A = A \quad \text{and} \quad A I_n = A.$$

IF-3 Find all 2×2 matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

such that

$$A^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Vector and Matrices

IF-4 Show that matrix multiplication is not commutative by calculating $AB - BA$.

(a)

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

(b)

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ -1 & 2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 & -2 \\ 3 & -2 & 4 \\ -3 & 5 & -1 \end{pmatrix}$$

Matrix Algebra

IF-5

(a) Let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

Compute A^2, A^3.

(b) Find A^2, A^3, A^n if

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

IF-6

Let A, A', B, B' be 2×2 matrices and O the 2×2 zero matrix. Express in terms of these five matrices the product of the 4×4 matrices

$$\begin{pmatrix} A & O \\ O & B \end{pmatrix} \begin{pmatrix} A' & O \\ O & B' \end{pmatrix}.$$

IF-7

Let

$$A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}.$$

Show there are no values of a and b such that

$$AB - BA = I_2.$$

IF-8

(a) If

$$A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

what is the 3×3 matrix A?

(b) If

$$A \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \\ 1 \end{pmatrix},$$

what is A?

IF-9

A square $n \times n$ matrix is called orthogonal if

$$A^T = I_n.$$

Show that this condition is equivalent to saying that:

(a) Each row of A is a vector of length 1.

(b) Two different rows are orthogonal vectors.

Matrix Algebra

IF-10

Suppose A is a 2×2 orthogonal matrix whose first entry is $a_{11} = \cos\theta$. Fill in the rest of A (there are four possibilities). Use Exercise 9.

IF-11

Show that if $A + B$ and AB are defined, then

$$(A+B)^T = A^T + B^T,$$
$$(AB)^T = B^T A^T.$$

IG: Solving Square System, Inverse Matrices

For each of the following, solve the equation $Ax = b$ by finding A^{-1}.

IG-1

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & -1 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 8 \\ 3 \\ 0 \end{pmatrix}.$$

IG-2

(a)

$$A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

IG-3

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}.$$

IG-4

Referring to Exercise 3 above, solve the system

$$x_1 - x_2 + x_3 = y_1,$$
$$2x_2 + x_3 = y_2,$$

$$-x_1 - x_2 + 2x_3 = y_3,$$

for the x_i as functions of the y_i.

16–5 Show that $(AB)^{-1} = B^{-1}A^{-1}$ by using the definition of inverse matrix.

16–6 Another calculation of the inverse matrix.

If we know A^{-1}, we can solve the system

$$AX = Y$$

for X by writing

$$X = A^{-1}Y.$$

But conversely, if we can solve by some other method (elimination) for X in terms of Y, getting

$$X = BY,$$

then the matrix $B = A^{-1}$ and we will have found A^{-1}.

This is a good method if A is an upper or lower triangular matrix — one with only zeros respectively below or above the main diagonal.

To illustrate:

(a) Let

$$A = \begin{pmatrix} -1 & 1 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find A^{-1} by solving

$$\begin{aligned} -x_1 + x_2 + 3x_3 &= y_1, \\ 2x_2 - x_3 &= y_2, \\ x_3 &= y_3, \end{aligned}$$

for the x_i in terms of the y_i (start from the bottom and proceed upwards).

(b) Calculate A^{-1} by the method given in the notes.

16–7 Consider the rotation matrix

$$A_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

corresponding to rotation of the x and y axes through the angle θ. Calculate A_θ^{-1} by the adjoint matrix method, and explain why your answer looks the way it does.

16–8

(a) Show: A is an orthogonal matrix (cf. Exercise 1F–9) if and only if

$$A^{-1} = A^T.$$

(b) Illustrate with the matrix of Exercise 7 above.

(c) Use (a) to show that if A and B are $n \times n$ orthogonal matrices, so is AB.

16–9

(a) Let A be a 3×3 matrix such that $|A| \neq 0$. The notes construct a right-inverse A^{-1}, that is, a matrix such that

$$AA^{-1} = I.$$

Show that every such matrix A also has a left inverse B (i.e. a matrix such that $BA = I$).

Hint: Consider the equation

$$A^{-1}(AA^{-1}) = IA^{-1}.$$

(b) Deduce that $B = A^{-1}$ by a non-line argument.